

LUMPREM2

Version 2 of the LUMPREM Groundwater Recharge Model

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April, 2021

Table of Contents

1. Introduction	1
1.1 General.....	1
1.2 LUMPREM2 Output Files.....	1
1.2.1 Tabular Output File	1
1.2.2 CSV File.....	1
1.2.3 Site Sample File	2
1.2.4 Run Record File	2
1.3 Running LUMPREM2	3
2. Concepts.....	4
2.1 Soil Moisture Stores	4
2.2 Operation of the Upper Soil Moisture Store.....	5
2.3 Operation of the Lower Soil Moisture Store.....	5
2.3.1 Drainage	5
2.3.2 Overflow.....	5
2.3.3. Extraction	5
2.3.4 Volume to Elevation.....	6
3. The LUMPREM2 Input File	7
3.1 Backwards Combability.....	7
3.2 Units	7
3.3 File Specifications.....	7
3.3.1 General.....	7
3.3.2 Specifics.....	7

1. Introduction

1.1 General

LUMPREM2 extends the functionality of LUMPREM. It is backwards compatible with LUMPREM. Hence it can be used with existing LUMPREM input files. However its main output file is slightly different from that of LUMPREM. It also records some new output files that the original version of LUMPREM does not provide.

For the moment, LUMPREM2 is offered as a separate program. The manual for the original version of LUMPREM should be read prior to reading the manual for LUMPREM2 that you are reading now.

Features that are new to LUMPREM2 include the following:

- Error checking of LUMPREM2 input files is greatly improved over that of LUMPREM.
- Blank lines and comments are allowed in LUMPREM2 input files.
- LUMPREM2 writes a brief run record file in which the values of its input variables are echoed. A user can therefore verify that these are as expected.
- Optionally, LUMPREM2 records a CSV output file in addition to its tabular output file. The contents of this file are almost identical to those of its tabular output file.
- Optionally, one or a number of time series produced by LUMPREM can be recorded in a site sample file (also known as a bore sample file). This type of file is easily read, and its contents readily manipulated, by members of the PEST Groundwater Utility suite.
- LUMPREM2 features two soil moisture stores instead of one. The lower store is referred to as a “below root zone” soil moisture store. This store can delay and dampen recharge that emerges from the upper soil moisture store. The upper soil moisture store simulates root zone processes.

LUMPREM2 output files are now discussed in greater detail.

1.2 LUMPREM2 Output Files

1.2.1 Tabular Output File

As is discussed at length in documentation of LUMPREM, as LUMPREM calculates the moisture content of its single soil moisture store, it records this, as well as a suite of other quantities, in a tabular data file. The final line of this file contains totals. This file can be easily imported into a spreadsheet wherein its contents can be plotted after removal of the final line. The first line of this file contains column headers. The first of these headers is “days”; the first column lists simulation days, starting at zero.

LUMPREM2 writes a similar file. However this file contains extra columns that accommodate the presence of LUMPREM2’s extra moisture store. The headers on some of the columns that are retained from the original LUMPREM tabular output file are altered to differentiate between the upper and lower stores.

The LR2SERIES program that accompanies LUMPREM can read this file.

1.2.2 CSV File

If asked to do so through its command line, LUMPREM2 writes a CSV file as well as its normal tabular data file. The contents of this CSV file are identical to those of its tabular data output file except for omission of the last “totals” line. This file is immediately readable by a spreadsheet or plotting program; plots of the contents of selected columns against time (the first columns) are readily made.

1.2.3 Site Sample File

If asked to do so, LUMPREM2 records a site sample file (also known as a bore sample file). The format of this type of file is discussed in detail in Part A of the manual of the PEST Groundwater Utilities. A site sample file has four columns. The first of these columns contains site names. The second column contains dates, while the third column contains times. Data values are recorded in the fourth column. The following rules apply.

1. Site names are 20 characters or less in length. They must not contain a space. They must not begin with “#”.
2. Dates are expressed using either the *dd/mm/yyyy* or *mm/dd/yyyy* protocols.
3. All entries for a single site appear on neighbouring lines.
4. Entries for each site must be arranged in increasing temporal order.

The integrity of a site sample file can be checked using the SMPCHK utility supplied with the PEST Groundwater Utility suite. A file named *settings.fig* must reside in the working folder from which this, or other programs of the Groundwater Utility suite, are run. The contents *settings.fig* inform them of the date protocol employed by site sample and other files. See documentation of the Groundwater Utilities for further details.

To instruct LUMPREM2 to record a site sample file, include a “ssf file” section at the end of its input file. An example appears in Figure 3.2. Variables which appear in this section are defined in Table 3.1. They are also discussed below.

The “ssf file” section is optional. If present in a LUMPREM2 input file, it must be its last section.

The first data line within the “ssf file” section of a LUMPREM2 input file must be the name of the site sample file that LUMPREM2 must write.

The following line should contain three entries. The first entry must be either “dd/mm/yyyy” or “mm/dd/yyyy”. This informs LUMPREM2 of the protocol that it must use to record dates in the site sample file which it writes. Then, on the same line, must follow the date and time at which the LUMPREM2 simulation begins. The format used to express the date must be the same as that of which LUMPREM2 was previously advised. The simulation starting time must be expressed using the format *hh:min:ss*. This time will comprise all entries in the “time” column of the site sample file that LUMPREM2 writes.

The following line of the LUMPREM2 input file must hold a single integer. This informs LUMPREM2 of the number of lines to follow. As presently coded, this integer must be a minimum of 0 and a maximum of 20. If it is 0, then a site sample file is not written.

Each of the following lines of the LUMPREM2 input file has 6 entries. The first entry on each line is the name of a LUMPREM2 output type. This name must be in exact accordance with the name of a column in its tabular output file and CSV output file. Then follows the name that LUMPREM2 will give to these same data in the site sample file which it writes. You get to choose this name yourself; it must be 15 characters or less in length and contain no spaces. The next two entries are the scale and offset that are applied to LUMPREM2 outputs of this type before they are recorded in the site sample file; first LUMPREM2 multiplies by the scale; then it adds the offset. Finally, lower and upper bounds are provided; these are enforced on the scaled and offset quantities recorded by LUMPREM2.

1.2.4 Run Record File

After it has read all of its input data, and before it undertakes any calculations, LUMPREM2 records all data that it reads from its input file in another file. This enables you to verify that it has read the contents of this file correctly. Unlike its other output files, this file always has the same name; it is called *lumprem_variables.rec*.

1.3 Running LUMPREM2

LUMPREM2 can be run in either of two ways.

The first alternative is to supply the names of its input and output files in its command line. In this case it is run using the command:

```
LUMPREM2 infile tabfile [csvfile]
```

Infile is the name of its input file. Specifications of this file are provided in the next section. *Tabfile* is name of its tabular output file. Optionally, the final name on the command line can be that of its CSV output file. If this name is missing, then LUMPREM2 does not write a CSV file.

The second option for running LUMPREM2 is to type its name at the screen prompt. Then LUMPREM2 asks for the names of these files itself. Prompts are as follows:

```
Enter name of LUMPREM input file: temp.in
Enter name for LUMPREM tabular output file: temp.dat
Enter name for CSV output file (<Enter> if none):
```

As it runs, LUMPREM2 informs a user of occasions (if any) when its solution procedure does not converge. It writes messages such as the following to the screen.

```
RECHMOD lower bucket: call 15 day 22 step 1: iteration limit exceeded
```

If you receive this message too many times, then consider increasing the number of time steps per day (NSTEP) and/or the number of iterations per time step (MXITER). Otherwise, do not worry too much about it.

2. Concepts

2.1 Soil Moisture Stores

Figure 2.1 depicts the conceptual model on which the LUMPREM2 numerical model is based.

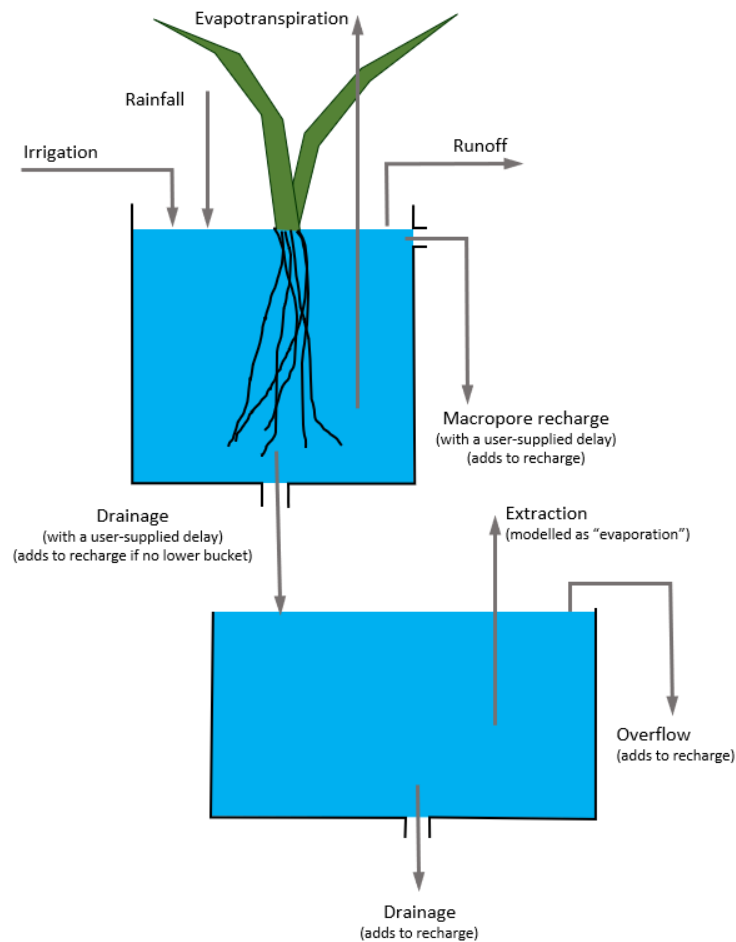


Figure 2.1. The LUMPREM2 conceptual model.

As is apparent from Figure 2.1, LUMPREM2 has two stores (sometimes referred to as “buckets”). The upper store is identical to that employed by the original LUMPREM simulator. This simulates processes in the plant root zone. The lower store simulates processes below the plant root zone and above the water table. Use of this lower bucket is optional.

If the lower store is, in fact, deployed, then drainage from the upper store is directed to the lower store. Optionally, drainage from the upper store can experience a delay before being provided to the lower store. Processes that operate in the lower store are similar to those that operate in the upper store, but are somewhat simpler.

When only a single soil moisture store is operative, groundwater recharge is calculated as drainage from this store (i.e. the upper store) plus macropore recharge. Both are subject to user-specified delays before being declared as “groundwater recharge”. When two moisture stores are operative, groundwater recharge is calculated as the sum of drainage from the lower store, overflow from the lower store and macropore recharge from the upper store. Only the last of these suffers a user-specified delay. Note, however, that drainage from the upper store is subject to a user-specified delay before being supplied to the lower store.

2.2 Operation of the Upper Soil Moisture Store

Operation of the upper soil moisture store in LUMPREM2 is identical to its operation in LUMPREM.

2.3 Operation of the Lower Soil Moisture Store

The lower soil moisture store receives water that drains from the upper soil moisture store. As stated above, this is optionally delayed before being supplied to the lower store. The latter store can lose water in one of three ways, these being:

1. drainage;
2. overflow;
3. extraction.

2.3.1 Drainage

Drainage from the lower soil moisture store is calculated in the same way as drainage from the upper soil moisture store. The equation is:

$$D = K_s [v']^l \left[1 - (1 - [v']^{1/m})^m \right]^2 \quad (2.1)$$

where:

D = rate of drainage;

K_s = saturated hydraulic conductivity;

v' = relative volume of water in the soil moisture store (i.e. v/v_{max});

l = pore-connectivity parameter (normally about 0.5); and

m = a parameter determining the shape of the drainage rate vs. stored water relationship (see documentation of LUMPREM).

Obviously, the values of the above parameters (including v_{max}) for the lower store can be different from those of the upper store. Equation 2.1 is graphed in documentation of LUMPREM.

2.3.2 Overflow

The lower store overflows when its volume exceeds v_{tot} , calculated as:

$$v_{tot} = v_{max} + v_{ext} \quad (2.2)$$

The use of this “extra volume” (i.e. v_{ext}) allows a user to specify an overflow volume that exceeds the maximum volume (i.e. v_{max}) that is used in equation 2.1.

2.3.3. Extraction

Extraction from the lower store is calculated in the same way as evaporative extraction is calculated for the upper store. However this extraction is “evaporation” in name only. Its status as “pseudo evaporation” allows the rate of water extraction from this store to be reduced as the volume of water in the store decreases. The equation that is used to enforce this reduction is exactly the same as that used in the upper store. The equation is:

$$E = f E_p \frac{1 - e^{-\gamma v'}}{1 - 2e^{-\gamma} + e^{-\gamma v'}} \quad (2.3)$$

where:

E = rate of water loss;

E_p = “potential evapotranspiration rate”;

f = “crop factor”;

v' = relative volume of water in the lower soil moisture store (i.e. v/v_{max} where v is the current volume of water in the store and v_{max} is the volume of the lower soil moisture store); and

γ = a parameter determining the shape of the evaporation rate vs. stored water relationship (see documentation of LUMPREM).

The “crop factor” for the lower soil moisture store is internally set to 1.0. A time series of “potential evaporation” is supplied by the user; this is a different time series from that applied to the upper moisture store. So too is the value of γ .

Equation 2.3 is graphed in the LUMPREM manual.

As stated above, extraction from the lower store is “evaporation” in name only. When LUMPREM2 calculates potential evaporation available for groundwater, it does not subtract water lost from the lower soil moisture store from potential evaporation applied to the upper store. It subtracts only evaporative losses from the upper store in order to calculate potential evaporation available to groundwater.

Simulation of water loss from the lower store as a kind of pseudo evapotranspiration is included in LUMPREM2 functionality simply as means of providing a little extra flexibility when LUMPREM2 is deployed to calculate heads or water depths at groundwater model boundaries. This is now discussed.

2.3.4 Volume to Elevation

The original version of LUMPREM allows a user to calculate a notional elevation from volume of stored water using the following equation.

$$h = a + f_1 v + f_2 v^p \quad (2.4)$$

where:

h is the calculated elevation;

v is the volume of water in the store;

a is an offset;

f_1 is a factor (referred to as “factor 1” in LUMPREM input);

f_2 is a factor (referred to as “factor 2” in LUMPREM input); and

p is a power.

As well as elevations, LUMPREM also records depths. These are calculated by subtracting water elevations from a reference elevation (for example the elevation of a topographic surface); the latter is supplied by the user. Let the reference (topographic) elevation be denoted by s . Depth to water d is calculated using the formula:

$$d = s - h \quad (2.5)$$

where:

d is the depth to water; and

s is the reference elevation.

LUMPREM2 uses these same formulae to convert volumes of water contained in a soil moisture store to elevations and depths. When using LUMPREM2, the user chooses which store is used for this purpose. (For the original LUMPREM it is the upper store, because this is the only store which it employs.)

3. The LUMPREM2 Input File

3.1 Backwards Combability

In order to maintain compatibility with the original version of LUMPREM, some aspects of the LUMPREM2 input file are optional. Hence a LUMPREM input file can serve as an input file for LUMPREM2. Of course, a file that serves as an input file for the original version of LUMPREM does not invoke a lower soil moisture store.

As was discussed in the previous section, operation of the lower and upper stores of a LUMPREM2 model are governed by similar equations. Variables that govern operation of LUMPREM2's lower store have similar names to those that govern operation of its upper store. The former names are derived from the latter names by affixing the prefix "_br" to these names; "br" stands for "below root zone".

3.2 Units

As for LUMPREM, LUMPREM2's time units are days. Length (and volume) units are immaterial; LUMPREM2's only requirement is that they be consistent.

3.3 File Specifications

3.3.1 General

Like a LUMPREM input file, a LUMPREM2 input file is divided into sections. The name of each section begins with the character "*". This is followed by a space, followed by the name of the section.

LUMPREM2 allows any line of its input file to be blank. On any line, any characters which follow a "#" character are ignored. Comments can therefore be included in a LUMPREM2 input file.

3.3.2 Specifics

LUMPREM2 input variables are recognized by their position in the file. These are set out in Figure 3.1. Square brackets denote optional variables in this figure. Figure 3.2 shows an example of a LUMPREM2 input file. Table 3.1 describes all variables.

```

* earth properties
maxvol irrigvolfrac [maxvol_br  extravol_br  gamma_br]
rdelay mdelay
ks m l mflowmax  [ks_br m_br l_br]
* volume to elevation
offset factor1 factor2 power [elevmin elevmax]  [upper_or_lower]
* topographic surface
surface
* initial conditions
vol [vol_br]
nrbuf nmbuf
(rbuf(i), i=1,nrbuf)
(mbuf(i), i=1,nmbuf)
* solution parameters
nstep mxiter tol
* timing information
numdays noutdays
(outday(i),i=1,noutdays)
* data filenames
vegfile OR crofac_all  gamma_all
rainfile
epotfile
irrigfile OR irrigcode_all  gwirrigfrac_all
[epotfile_br] OR [epot_br_all]
* ssf file
ssffile
date_spec  starting_date starting_time
numssfseries
lumprem_series_name ssf_series_name  scale_ssf offset_ssf minval_ssf  maxval_ssf
(the above line should appear numssfseries times)

```

Figure 3.1. Structure of a LUMPREM2 input file. Variables in square brackets are optional. The “ssf file” section is optional.

```

* earth properties
# maxvol irrigvolfrac maxvol_br extravol_br gamma_br
1.0 0.5 0.5 0.1 2.0
# rdelay mdelay
4.5 1.5
# ks m l mflowmax ks_br m_br l_br
0.1 0.5 0.5 0.1 1.0e-3 0.5 0.5

* volume to elevation
# offset factor1 factor2 power elevmin elevmax lower_or_upper
0.0 2.0 3.0 0.5 -1.0e30 1.0e30 lower

* topographic surface
# surface
0.0

* initial conditions
# vol vol_br
0.5 0.0
# nrbuf #nmbuf
1 2
# rbuf()
0.0
# mbuf()
0.0 0.1

* solution parameters
# nstep mxiter tol
20 200 1e-05

* timing information
# numdays noutdays
3653 120
31 60 91 121 152 182 213 244 274 305 335 366 397 425 456 486 517 547 578 609 639
670 700 731 762 790 821 851 882 912 943 974 1004 1035 1065 1096 1127 1155 1186 1216
1247 1277 1308 1339 1369 1400 1430 1461 1492 1521 1552 1582 1613 1643 1674 1705
1735 1766 1796 1827 1858 1886 1917 1947 1978 2008 2039 2070 2100 2131 2161 2192
2223 2251 2282 2312 2343 2373 2404 2435 2465 2496 2526 2557 2588 2616 2647 2677
2708 2738 2769 2800 2830 2861 2891 2922 2953 2982 3013 3043 3074 3104 3135 3166
3196 3227 3257 3288 3319 3347
3378 3408 3439 3469 3500 3531 3561 3592 3622 3653

* data filenames
# vegfile
veg.dat
# rainfile
rain.dat
# epotfile
epot.dat
# irrigfile
0 0.0
# epotfile_br
0.0

* ssf file
# ssffile
temp.ssf
# date_format starting_date starting_time
dd/mm/yyyy 1/1/2000 00:00:00
# numssfser
4
# lumpsername ssfsername scale_ssf offset_ssf minval_ssf maxval_ssf
total_rech recharge 1.0 0.0 -1e30 1e30
drain_upper soildraing 1.0 0.0 -1e30 1e30
net_recharge net_recharge 1.0 0.0 -1e30 1e30
depth-to-water depth 1.0 0.0 -1e30 1e30

```

Figure 3.2 An example of a LUMPREM2 input file.

Variables appearing in a LUMPREM2 input file are listed in Table 3.1. Variables are real numbers, unless stated otherwise. For more details see documentation of LUMPREM.

Table 3.1 Variables which appear in a LUMPREM2 input file. Variables are real numbers unless stated otherwise.

Variable	Role
<i>maxvol</i>	The volume of the upper soil moisture store. This must be greater than zero.
<i>irrigvolfrac</i>	The fractional soil moisture volume which irrigation water must maintain. See documentation of LUMPREM for more details.
<i>maxvol_br</i>	The volume of the lower soil moisture store. Set this to zero (or omit this variable altogether) to disable the lower soil moisture store.
<i>extravol_br</i>	The rate of drainage of moisture from the lower soil moisture store is calculated using v/v_{max} , where v is the current moisture volume and v_{max} is the maximum soil moisture volume (i.e. input variable <i>maxvol_br</i>). However, the lower soil moisture store does not overflow until it reaches a level of <i>maxvol_br</i> + <i>extravol_br</i> . <i>Extravol_br</i> must be set to zero or greater.
<i>gamma_br</i>	This is γ of equation 2.3 applied to the lower LUMPREM2 store. In contrast to the upper store, there is no option for this variable to vary with time.
<i>rdelay</i>	The delay, in days, between water draining from the upper soil moisture store (calculated using equation 2.1) and being assigned to either groundwater recharge (if the lower soil moisture store is absent) or as inflow to the lower soil moisture store. This is a real number; it can include fractional days.
<i>mdelay</i>	The delay, in days, between water leaving the upper soil moisture store (calculated as overflow of this store up to a maximum rate of <i>mflowmax</i>) and being assigned to macropore groundwater recharge. This is a real number; it can include fractional days.
<i>ks</i>	K_s of equation 2.1 as supplied to the upper soil moisture store.
<i>m</i>	m of equation 2.1 as supplied to the upper soil moisture store.
<i>l</i>	l of equation 2.1 as supplied to the upper soil moisture store.
<i>mflowmax</i>	Maximum macropore recharge allowed per day.
<i>ks_br</i>	K_s of equation 2.1 as supplied to the lower soil moisture store.
<i>m_br</i>	m of equation 2.1 as supplied to the lower soil moisture store.
<i>l_br</i>	l of equation 2.1 as supplied to the lower soil moisture store.

<i>offset</i>	<i>a</i> of equation 2.4.
<i>factor1</i>	f_1 of equation 2.4.
<i>factor2</i>	f_2 of equation 2.4.
<i>power</i>	p of equation 2.4.
<i>elevmin</i>	Outcomes of equation 2.4 are clipped at this value if they are below it. This variable is optional.
<i>elevmax</i>	Outcomes of equation 2.4 are clipped at this value if they are above it. This variable is optional.
<i>lower_or_upper</i>	Informs LUMPREM2 whether equation 2.4 applies to moisture stored in the upper or lower stores. The value supplied for this variable must be either “lower” or “upper”.
<i>vol</i>	The initial volume of water in the upper soil moisture store.
<i>vol_br</i>	The initial volume of water in the lower soil moisture store. This variable can be omitted if <i>maxvol_br</i> is not supplied, or is supplied as 0.0.
<i>nrbuf</i>	The number of elements of the <i>rbuf</i> array to be supplied on the next line. This is an integer variable.
<i>nmbuf</i>	The number of elements of the <i>mbuf</i> array to be supplied two lines down. This is an integer variable.
<i>(rbuf(i), i=1,nrbuf)</i>	The recharge delay buffer. Supply initial values for the elements of this array. The content of the first element is assumed to have left the soil moisture store on the previous day; the content of the second element is assumed to have drained from the soil moisture store two days previously, etc. All elements of this array must be recorded on a single line of the LUMPREM2 input file.
<i>(mbuf(i), i=1,nmbuf)</i>	The macropore recharge delay buffer. Supply initial values for the elements of this array. The content of the first element is assumed to have left the soil moisture store on the previous day; the content of the second element is assumed to have left the soil moisture store two days earlier, etc. All elements of this array must be recorded on a single line.
<i>nstep</i>	The number of steps into which each day is subdivided for the purpose of iterative soil moisture store computation. This is an integer variable.
<i>mxiter</i>	The maximum number of iterations per time step. LUMPREM2 accepts the calculated value of soil moisture store after this many iterations have elapsed, regardless of whether the solution process has converged to the desired tolerance. This is an integer variable.

<i>tol</i>	The convergence tolerance. The solution is assumed to have converged when two successive estimates of v' differ by less than <i>tol</i> .
<i>numdays</i>	The total number of days in a LUMPREM2 simulation. This is an integer variable.
<i>noutdays</i>	The number of days for which LUMPREM2 output is required (equivalent to the number of times that LUMPREM2 calls the RECHMOD subroutine – see documentation of the original version of LUMPREM). This is an integer variable.
(<i>outday(i)</i> , <i>i=1,noutdays</i>)	An integer array with <i>noutdays</i> elements listing the simulation days at which output is required. If desired, elements of this array can wrap onto ensuing lines.
<i>vegfile</i>	The name of the “vegetation file” which LUMPREM2 must read to obtain crop factor and γ data. Alternatively, time-invariant values of the crop factor and gamma (in that order) can be read from this line. See documentation of LUMPREM for the format of the vegetation file.
<i>rainfile</i>	The name of the “rainfall file” which LUMPREM2 must read to obtain rainfall data. See documentation of LUMPREM for the format of this file.
<i>epotfile</i>	The name of the “evaporation file” which LUMPREM2 must read to obtain potential evaporation data. The contents of this file are applied to the upper soil moisture store. See documentation of LUMPREM for the format of this file.
<i>irrigfile</i>	The name of the “irrigation file” which LUMPREM2 must read to obtain irrigation data. Alternatively, time-invariant values for the irrigation code and groundwater extraction fraction (in that order) can be read from this line. The irrigation code is an integer (zero or one), while groundwater extraction fraction is a real number. See documentation of LUMPREM for the format of the irrigation file.
<i>epotfile_br</i>	If <i>maxvol_br</i> is greater than 0.0 (indicating the presence of a lower store), then either the name of a “potential evaporation” file must be supplied, or a single value for potential evaporation must be supplied (i.e. the variable <i>epot_br_all</i>). If <i>epot_br_all</i> (a real number) is supplied, then this rate of potential evaporation is applied during every simulation day; often this rate is 0.0. If supplied, file <i>epotfile_br</i> employs the same format as <i>epotfile</i> for recording of daily rates of potential evaporation. See the LUMPREM manual for details.
<i>ssffile</i>	The name of the site sample file which LUMPREM2 must write. This is supplied only if a “site sample file” section is present in the LUMPREM2 input file.
<i>date_format</i>	This must be supplied as either “dd/mm/yyyy” or “mm/dd/yyyy”. It determines the protocol which LUMPREM2 must use to express dates in its site sample output file.

<i>starting_date</i>	The date at which the LUMPREM2 simulation begins. Supply this date using the protocol provided with <i>date_format</i> .
<i>starting_time</i>	The time of day at which the LUMPREM2 simulation begins. This time will comprise the third entry of all lines of the LUMPREM2-produced site sample file. The protocol is <i>hh:mm:ss</i> .
<i>numssfser</i>	The number of time series to record in the site sample file; hence the number of lines to follow. This is an integer.
<i>lumpsername</i>	Provide the name of a LUMPREM2-generated time series. Time series names comprise the headers to columns appearing on LUMPREM2's tabular and CSV output files. They must be provided exactly as recorded by LUMPREM2. Quotes around these names are optional.
<i>ssfsername</i>	This is the "site name" that will be attributed to the time series in the site sample file which LUMPREM2 records. LUMPREM2 insists that this name be 15 characters or less in length, and does not contain a space. Optionally it can be surrounded by quotes.
<i>scale_ssf</i>	Each element of a LUMPREM2-calculated time series is multiplied by <i>scale_ssf</i> and then summed with <i>offset_ssf</i> before being recorded in the site sample file.
<i>offset_ssf</i>	Each element of a LUMPREM2-calculated time series is multiplied by <i>scale_ssf</i> and then summed with <i>offset_ssf</i> before being recording in the site sample file.
<i>minval_ssf</i>	LUMPREM2 clips time series elements from below at this number before recording them in its site sample output file.
<i>maxval_ssf</i>	LUMPREM2 clips time series elements from above at this number before recording them in its site sample output file.